

DR-AGENTOS: PREDICTIVE 4D BELIEF FOR NAVIGATION IN EVOLVING WORLDS

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ABSTRACT

Embodied navigation is predominantly defined against a stationary world snapshot: maps and memories are treated as descriptions of the state in which the agent will act. This premise fails for long-lived agents because the world continues to evolve while it is unobserved. We identify and formulate *persistent navigation in evolving worlds*, a new problem in which an agent must infer the current task-relevant state from partial, irregular, and temporally stale experience, navigate under that uncertainty, and revise its decisions as physical evidence arrives. Unlike dynamic mapping that reacts to observed changes or navigation that tracks an already moving target, this setting centers on *hidden evolution between visits*: the memory can be historically correct yet operationally obsolete. We introduce DR-AGENTOS, a predictive embodied-agent architecture whose P4D-NAV core converts an evidence-grounded 4D memory into an actionable belief over the present world. A continuous-time Transformer models irregular histories, and a structured persistence–relocation pointer head predicts whether the last supported state still holds and where the state likely transitioned otherwise. The agent uses the resulting distribution to choose between direct navigation and cost-aware inspection, then applies a visibility-aware Bayesian update to positive and negative observations before replanning. We also introduce P4D-BENCH, a 54-scene benchmark comprising 803.68k tasks across state prediction, object navigation, vision-and-language navigation, and embodied question answering. Paired static, routine, and random worlds test whether an agent learns evolution rather than exploiting scene-frequency shortcuts. Together, our formulation shifts embodied navigation from acting on a remembered past to maintaining, testing, and acting on beliefs about the world as it is now.

1 INTRODUCTION

Embodied navigation requires an agent to ground a goal, maintain a world model, and select actions from egocentric observations. Most formulations nevertheless adopt a snapshot abstraction: the map or memory is assumed to describe the same world state in which navigation is executed (Chaplot et al., 2020; Huang et al., 2023; Gu et al., 2024). Long-lived agents violate this assumption. Homes, offices, and campuses evolve while unobserved as objects are relocated, spaces are rearranged, and accessibility changes. The agent may therefore retain evidence that is historically correct yet decisionally stale, pointing to an obsolete goal or an inefficient route.

Long-term deployment creates a dual tension. A persistent agent is not a persistent observer: as its *4D history* grows, individual records become stale, yet repeated experience reveals environmental dynamics, routines, and personal habits. More history therefore does not automatically mean better knowledge of the present. Meanwhile, history grows without bound, whereas decision-time context and computation must remain controlled. Most embodied memories are retrospective, organizing or retrieving what was observed (Liu et al., 2024; Kurenkov et al., 2023); a predictive memory should instead infer *what task-relevant state is most likely true now, how uncertain that belief is, and how navigation should adapt*. We seek a decision-preserving interface in which the episodic store retains grounded facts while the action policy consumes a compact, task-conditioned belief that preserves prediction and search utility.

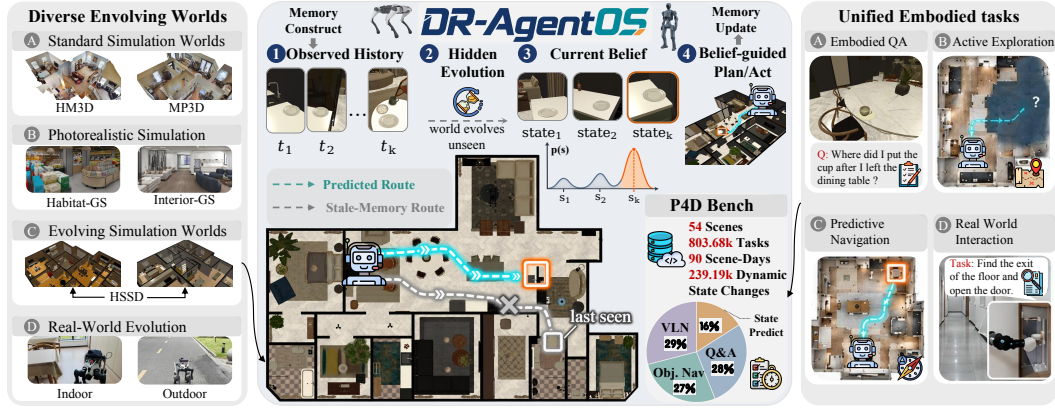


Figure 1: Overview of DR-AGENTOS in evolving worlds. Observed histories are transformed into current-state beliefs that guide planning and are updated by new evidence, supporting prediction, navigation, exploration, question answering, and real-world interaction.

Recent work models temporal persistence, future relations, household routines, and targets that move during navigation (Saavedra-Ruiz et al., 2026; Patel & Chernova, 2023; Argenziano et al., 2026; Dorbala et al., 2026). These approaches establish that dynamics matter, but leave a distinct problem insufficiently specified: navigation through changes that occur *between* observations. In this regime, neither a freshly updated map nor the trajectory of a currently observed target is available. The agent must infer the latent present from its own incomplete past and decide which physical action will most efficiently confirm or refute that inference.

Positioning against prior work. Our formulation separates four sources of difficulty that are often studied independently. Conventional ObjectNav and VLN primarily reason over an episode-level snapshot (Anderson et al., 2018; Savva et al., 2019; Chaplot et al., 2020); lifelong and dynamic mapping revise memory after a change is observed (Narayana et al., 2020; Liu et al., 2024); predictive scene models forecast future relations or placements, commonly as an open-loop prediction or ranked hypothesis set (Saavedra-Ruiz et al., 2026; Argenziano et al., 2026); and moving-target navigation plans against motion during execution (Dorbala et al., 2026). We instead study an agent that returns after an observation gap, faces a world whose task-relevant state may have changed unseen, and must close the loop between a calibrated current-state belief and evidence-seeking navigation. The key object is therefore not a more recent map or a single predicted destination, but an *action-facing belief over the present* that is repeatedly tested and revised. This positioning defines persistent navigation in evolving worlds as a problem at the intersection of long-term memory, temporal prediction, active perception, and embodied planning.

As illustrated in Figure 1, DR-AGENTOS places this predictive loop in a unified embodied system. It constructs memory from conventional simulation (HM3D and MP3D), high-fidelity simulation (Habitat-GS and Interior-GS), dynamic HSSD scenes, and indoor or outdoor real-world observations. When the world evolves out of view, the agent transforms observed history into a belief over the current state, uses that belief to plan and act rather than blindly following a stale-memory route, and writes newly collected evidence back to memory. The same interface supports embodied question answering, active exploration, predictive navigation, and real-world interaction. P4D-BENCH provides the controlled simulated evaluation, while the real-world panel illustrates the deployment interface rather than a benchmark result.

We formulate this broader setting as **persistent navigation in evolving worlds**: before a query, an agent has accumulated timestamped observations through intermittent visits, while the task-relevant world state may have changed during the resulting observation gaps. We operationalize its central uncertainty through **persistent dynamic object navigation**, where a portable target may undergo hidden transitions after its last positive observation. This controlled instantiation is deliberately narrower than the new problem: it makes the latent state, evidence, and navigation cost measurable without reducing the conceptual setting to object displacement alone. The agent receives neither the current target state nor the activity that caused the change and must navigate under a fixed budget

using only causal memory, candidate receptacles and viewpoints, and online RGB-D observations. Our main protocol freezes the target after the query to isolate reasoning over stale history; an extension allows another hidden transition during navigation.

We introduce DR-AGENTOS, a long-term embodied agent centered on **4D Predictive Memory**, and instantiate its predictive-navigation core as P4D-NAV. Its 4D store maintains persistent identities, time-valid states, and observation provenance in a common frame. A causal retriever selects target-relevant history, and a lightweight continuous-time Transformer represents irregular gaps and calendar periodicity. A structured head factorizes the current belief into the probability that the last-seen state persists and a pointer distribution over relocation candidates. This matches the navigator’s physical alternatives, supports variable candidate sets, and exposes the belief—rather than the event sequence—as the memory interface to the action policy.

Prediction is embedded in a closed decision loop. Concentrated beliefs support a direct attempt; diffuse beliefs induce probability–cost-aware exploration. Navigation returns new RGB-D evidence, and a visibility-aware Bayesian update suppresses an inspected hypothesis only when the target should have been observable there. The agent then replans from the posterior. The same controller handles rare, routine, personal, and irregular changes without an oracle dynamics label, connecting history $\rightarrow$ current-state belief $\rightarrow$ plan/action $\rightarrow$ evidence $\rightarrow$ memory update.

We evaluate this capability with P4D-BENCH, a 54-scene benchmark containing 803.68k task instances across state prediction, object navigation, vision-and-language navigation, and embodied question answering. Its evolving HSSD histories combine longitudinal activity records, real human–object interaction traces, and household state-transition programs. Our controlled analysis measures belief quality, navigation efficiency, evidence-aware recovery, and generalization. Paired static, routine, and random worlds share the observed setup while differing only in the hidden transition process, testing whether gains arise from learnable temporal structure rather than scene frequency or route geometry.

Contributions. Our contributions are threefold:

- We identify and formulate **persistent navigation in evolving worlds**, a new embodied problem in which unobserved evolution makes long-term history simultaneously stale evidence about the past and predictive evidence about the present; persistent dynamic object navigation provides a controlled operationalization.
- We introduce DR-AGENTOS and its P4D-NAV core with **4D Predictive Memory**, which transform causal history into an action-facing persistence–relocation belief and close the loop through physical evidence, memory update, and replanning.
- We introduce P4D-BENCH, a human-trace-grounded, 54-scene benchmark spanning state prediction, ObjectNav, VLN, and EQA with paired static/routine/random controls.

2 RELATED WORK

Embodied navigation and active search. Embodied navigation spans object-goal, language-conditioned, and exploratory tasks that couple semantic perception, spatial reasoning, and action (Anderson et al., 2018; Savva et al., 2019; Chaplot et al., 2020; Xin et al., 2026). Semantic maps and open-vocabulary scene representations improve generalization by associating language with metric or graph-structured space (Huang et al., 2023; Gu et al., 2024; Werby et al., 2024). Probabilistic object-search methods use object–room or object–object correlations to order exploration and update a belief after observations (Zheng et al., 2022; Ginting et al., 2024). These formulations generally assume that the task-relevant world state remains consistent with the map during an episode. P4D-NAV addresses a different source of uncertainty: historically valid observations can become decisionally stale because the environment evolves between visits.

Navigation in changing environments. Early work on dynamic mapping often filters moving content or updates a semantic map after changes are observed (Bescos et al., 2018; Narayana et al., 2020). Scene Graph Memory models a partially observable dynamic graph and predicts object–location links (Kurenkov et al., 2023), while DynaMem continually reconstructs an actionable spatio-semantic memory for mobile manipulation (Liu et al., 2024). Predictive spatio-temporal

scene graphs attach persistence and emergence models to open-vocabulary relations (Saavedra-Ruiz et al., 2026). FlowMaps learns multimodal continuous future object placements and uses ranked hypotheses for dynamic ObjectNav (Argenziano et al., 2026). Portable Object Navigation and Transit-Aware Planning model objects that move as the agent moves (Dorbala et al., 2026). Our setting is complementary: dynamics are inferred from the agent’s causal, intermittent history; object-specific regimes are hidden; and prediction is embedded in an observe–update–replan loop rather than consumed only as a fixed candidate ranking or known transition path.

Long-term embodied memory. Long-horizon benchmarks increasingly require agents to retain observations across exploration or interaction (Khanna et al., 2024b; Majumdar et al., 2024; Yadav et al., 2025; Zhang et al., 2026). GOAT-Bench evaluates multi-modal lifelong navigation in persistent but static scenes, whereas FindingDory evaluates recall and action over long interaction histories. Open-vocabulary 3D memories support spatial retrieval (Gu et al., 2024; Werby et al., 2024), while recent 4D representations index time-varying states or human-centered events (Sohn et al., 2025; Bartoli et al., 2026). These systems motivate persistent identity, temporal validity, and provenance. We use these properties as infrastructure, but focus the learning problem on a navigation-relevant belief about hidden evolution: whether an entity’s last state persists and, if not, which reachable state should be inspected next.

Human routines and personalized object dynamics. Household activities exhibit temporal and person-specific regularities. HOMER and SLATE-PRO predict routine object usage for proactive assistance (Patel & Chernova, 2023); STREAK studies continual adaptation under household context drift (Bartoli et al., 2024); and UCON evaluates navigation with a large user-habit knowledge base (Wang et al., 2026). Existing dynamic-world generators commonly rely on hand-designed routines or language-model priors. P4D-BENCH instead normalizes complementary longitudinal activity, real object-interaction, and complete state-transition records into executable HSSD episodes. Its paired worlds preserve marginal scene and query factors while changing the transition mechanism, enabling controlled measurement of when routine learning helps navigation.

3 PREDICTIVE 4D BELIEF NAVIGATION

3.1 PERSISTENT NAVIGATION IN EVOLVING WORLDS

At the system level, the task-relevant state may include object placement, spatial configuration, and accessibility. To isolate the coupling between hidden world evolution, predictive belief, and physical action, we formalize the core task through portable-object goals. Let an environment contain a navigable space $\mathcal{X}$, persistent entities $\mathcal{O}$, and candidate states $\mathcal{C}_o$ for each target $o \in \mathcal{O}$. A state identifies a semantically valid receptacle or region and its associated inspection viewpoints. Before a query at time t_q , the agent has collected only a partial history

$$\mathcal{H}_{\leq t_q} = \{(I_k, D_k, \xi_k, t_k) \mid t_k \leq t_q\}, \quad (1)$$

where I_k, D_k, ξ_k denote RGB, depth, and camera pose. The true query-time state $s_q^o \in \mathcal{C}_o \cup \{\text{unknown}\}$ and all transitions that occur outside the agent’s observations are hidden.

An episode supplies a target description, query time, initial pose x_0 , a causal memory materialized from $\mathcal{H}_{\leq t_q}$, and a finite action budget. At decision step j , the agent selects a candidate viewpoint or a low-level navigation action, receives observation z_j , and may terminate with STOP or NOT_FOUND. Success requires a correct stop near a valid goal viewpoint with the target visible. The policy should maximize success while minimizing travel and unnecessary inspections:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi}[\mathbb{I}(\text{success}) - \lambda_d L - \lambda_n N_{\text{inspect}}]. \quad (2)$$

The main protocol holds the target fixed after t_q , isolating whether stale history can be transformed into a useful belief about the present. An online-dynamic extension advances the world clock during execution and may apply an additional hidden transition outside the camera view.

3.2 FROM CAUSAL 4D MEMORY TO CURRENT-STATE BELIEF

Evidence-grounded 4D memory. The memory represents a persistent 3D semantic scene whose entity states evolve over time. For image coordinate $\tilde{\mathbf{u}} = [u, v, 1]^\top$ with depth $D_t(u, v)$, the

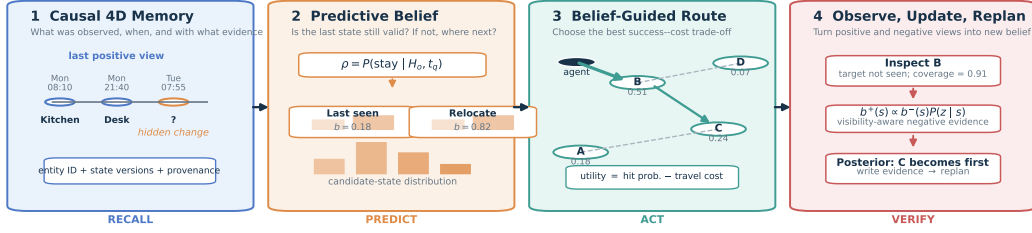


Figure 2: Predictive-navigation core of DR-AGENTOS. A causal 4D memory is propagated to the query time as a structured current-state belief, which the agent uses to plan, inspect, update its evidence, and replan.

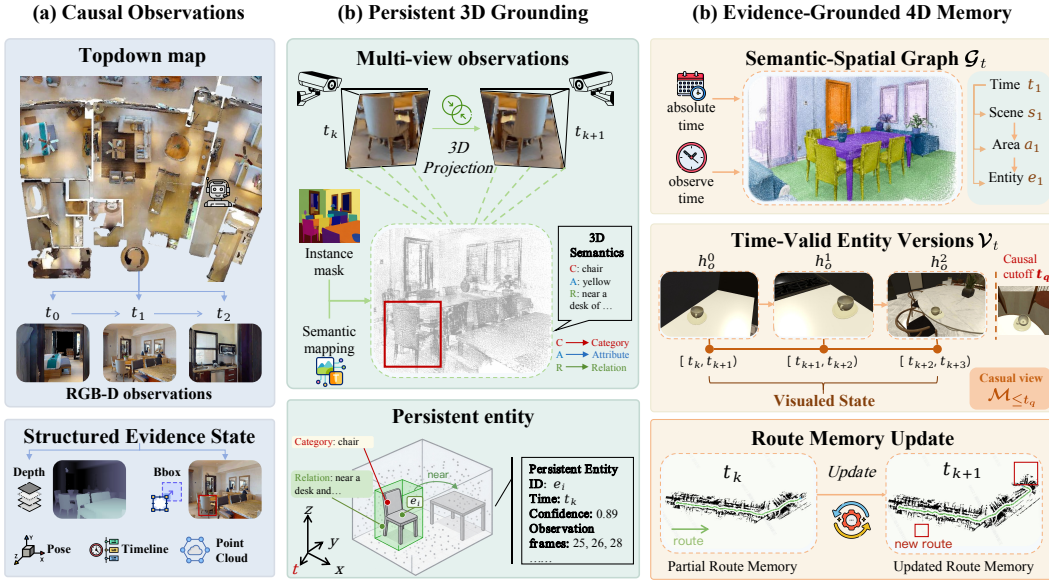


Figure 3: 4D memory construction. Multimodal observations are registered in a shared frame, consolidated into persistent entities, and retained as temporally versioned states with supporting evidence.

corresponding world point is

$$\mathbf{x}^w = \text{proj}_3 \left(\mathbf{T}_{w \leftarrow c, t} \begin{bmatrix} D_t(u, v) \mathbf{K}^{-1} \tilde{\mathbf{u}} \\ 1 \end{bmatrix} \right), \quad (3)$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\mathbf{T}_{w \leftarrow c, t}$ is the camera-to-world transform. Cross-view association consolidates observations using semantic compatibility, 3D overlap, temporal consistency, and source confidence. Memory is materialized causally from versioned deltas:

$$\mathcal{M}_{\leq t} = \text{Materialize}(\Delta \mathcal{M}_0, \dots, \Delta \mathcal{M}_t) = (\mathcal{G}_t, \mathcal{V}_t, \mathcal{B}_t), \quad (4)$$

where $\mathcal{G}_t$ stores semantic-spatial relations, $\mathcal{V}_t$ stores time-valid entity versions, and $\mathcal{B}_t$ stores observation provenance. The m -th version of entity o is

$$h_o^m = ([t_{o,s}^m, t_{o,e}^m], s_o^m, c_o^m, f_o^m, \mathcal{P}_o^m), \quad (5)$$

containing a validity interval, candidate state, confidence, visual feature, and evidence handles. Persistent identity is separated from mutable state: a change appends a version without erasing the entity's earlier trajectory, and every query-time view excludes evidence after t_q .

Target-conditioned history and continuous time. For target o , a typed memory query resolves its persistent identity and returns the most recent K causal events

$$H_o = \{e_k = (o_k, s_k, t_k, z_k, c_k, v_k)\}_{k=1}^K, \quad (6)$$

where s_k is the observed candidate state; $z_k \in \{\text{seen}, \text{notseen}\}$ is the evidence type; c_k is perception confidence; and v_k measures whether that state was adequately visible. Context events may share an entity, receptacle, region, or time window with the target, but hidden activities and transition labels are never exposed. Each scene-specific candidate is encoded from its context, functional role, geometry, and instance identity; an additional `unknown` candidate absorbs states outside the mapped set.

An event token combines entity, state, evidence, confidence, and visibility with elapsed and calendar time:

$$\mathbf{x}_k = E_o(o_k) + E_s(s_k) + E_z(z_k) + E_c(c_k) + E_v(v_k) + \phi_\Delta(t_q - t_k) + \phi_{\text{cal}}(t_k), \quad (7)$$

where ϕ_Δ includes $\log(1 + t_q - t_k)$ and ϕ_{cal} contains sinusoidal hour-of-day and day-of-week features. A learned query token specifies the target, query time, and last positive state. A compact Transformer yields

$$(\mathbf{h}_1, \dots, \mathbf{h}_K, \mathbf{h}_q) = \text{CTTransformer}(\mathbf{x}_1, \dots, \mathbf{x}_K, \mathbf{x}_q). \quad (8)$$

It therefore models irregular gaps, repeated negative evidence, and event order without an oracle mobility type or the activity responsible for a hidden change.

Structured persistence–relocation belief. A direct classifier entangles whether a state changed with where it changed to. We factorize these questions. Let s_{last} be the last positive state. The persistence branch predicts

$$\rho_q = P(s_q^o = s_{\text{last}} \mid H_o, t_q) = \sigma(\mathbf{w}_\rho^\top \mathbf{h}_q). \quad (9)$$

For each candidate $i \neq s_{\text{last}}$, a shared pointer head computes

$$a_i = \frac{(W_q \mathbf{h}_q)^\top (W_c \mathbf{g}_i)}{\sqrt{d}} + \psi(\mathbf{h}_q, \mathbf{g}_i), \quad \pi_q(i) = \frac{\exp a_i}{\sum_{j \neq s_{\text{last}}} \exp a_j}, \quad (10)$$

where $\mathbf{g}_i$ is the candidate representation. The prior belief is

$$b_q^-(i) = \begin{cases} \rho_q, & i = s_{\text{last}}, \\ (1 - \rho_q)\pi_q(i), & i \neq s_{\text{last}}. \end{cases} \quad (11)$$

The model is trained end-to-end with $\mathcal{L}_{\text{state}} = -\log b_q^-(s_q^o)$. Thus the factorization is an inductive bias rather than an oracle change label: it can preserve a strong last-state hypothesis, concentrate on recurring transitions, or expose diffuse uncertainty to the downstream agent.

3.3 PREDICTIVE EMBODIED AGENT

DR-AGENTOS treats the belief model as a tool inside an embodied decision loop, not as an open-loop destination predictor. Given a navigation or language goal, the agent (i) resolves the referenced entity and task constraints, (ii) queries the causal 4D memory, (iii) invokes P4D-NAV at the decision time, (iv) selects a navigation or inspection tool call from the returned belief and map costs, and (v) writes the resulting observation and posterior back to memory. A frozen high-level controller may express goals and tool calls in language, while the belief update and spatial interfaces remain typed and verifiable. The perception stack and low-level navigator are shared across all compared methods, so the learned component changes what the agent believes and where it gathers evidence, not its locomotion capability.

Belief-guided action. Each candidate i has reachable inspection viewpoints $\mathcal{W}_i$. Let $d(x, \mathcal{W}_i)$ be the shortest geodesic distance from the current pose and $\hat{v}_i$ the expected inspection coverage. The agent selects

$$i^* = \arg \max_{i \in \mathcal{C}_o \setminus \mathcal{I}} \frac{b(i)\hat{v}_i}{d(x, \mathcal{W}_i) + \lambda_{\text{inspect}}}, \quad (12)$$

where $\mathcal{I}$ contains sufficiently inspected hypotheses. A concentrated belief supports direct navigation; a diffuse belief induces cost-aware active inspection. The low-level navigator executes the chosen route and returns an RGB-D observation rather than a privileged success flag.

Evidence update and replanning. After inspecting candidate r , the agent obtains result z , detector reliability c , and effective visibility $v \in [0, 1]$. A positive instance match supports r ; a negative view should suppress it only when the target would have been observable:

$$P(z = \text{notseen} \mid s_q^o = i, c, v) = \begin{cases} 1 - cv, & i = r, \\ 1, & i \neq r. \end{cases} \quad (13)$$

The posterior is

$$b_q^+(i) = \frac{b_q^-(i)P(z \mid s_q^o = i, c, v)}{\sum_j b_q^-(j)P(z \mid s_q^o = j, c, v)}. \quad (14)$$

A clear empty view provides strong counter-evidence; an occluded view makes only a small change. The agent stores the inspection, visibility assessment, and posterior as a new causal memory event, removes only sufficiently tested hypotheses, and repeats the plan-act-observe cycle until it succeeds, exhausts the budget, or returns NOT_FOUND.

This single loop supports direct predictive navigation, belief-guided search, and recovery after an incorrect first destination. In online-dynamic episodes, the agent predicts at the candidate arrival time rather than only at departure:

$$b_{t+\hat{\tau}_i}^- = F_\theta(H_o, t + \hat{\tau}_i), \quad (15)$$

and replans after every action chunk. No explicit router chooses a “static” or “dynamic” policy; behavior emerges from the current belief, travel cost, and newly acquired evidence.

4 P4D-BENCH: EVALUATING NAVIGATION IN EVOLVING WORLDS

Existing navigation benchmarks typically expose a fixed scene or reset the environment between episodes, making the agent’s latest map a valid account of the world. They therefore cannot isolate the central difficulty of our setting: the world may change during an observation gap, while only the agent’s causal history is available at decision time. We introduce P4D-BENCH as a dedicated benchmark for this gap. The full release contains 54 scenes and 803.68k task instances over longitudinal hidden state changes. It couples continuous world evolution with executable embodied episodes so that the same latent state can be evaluated through prediction, navigation, language-conditioned behavior, and embodied question answering.

4.1 DESIGN PRINCIPLES AND SCOPE

P4D-BENCH is designed around four principles. First, *temporal continuity*: examples are sampled from persistent scene histories rather than independent object placements. Second, *causal observability*: the public memory contains only evidence that the robot could have acquired before the query. Third, *physical executability*: predicted states correspond to collision-free placements with reachable, visibility-checked inspection viewpoints. Fourth, *controlled dynamics*: paired worlds keep the scene, target, query, and marginal placement factors fixed while varying the hidden transition mechanism. These controls separate predictive reasoning from scene frequency, route geometry, and privileged simulator access.

The task pool comprises current-state prediction (16%), object navigation (27%), vision-and-language navigation (29%), and embodied question answering (28%). Prediction probes the belief independently of control. Navigation tests whether that belief produces useful destinations, search orders,

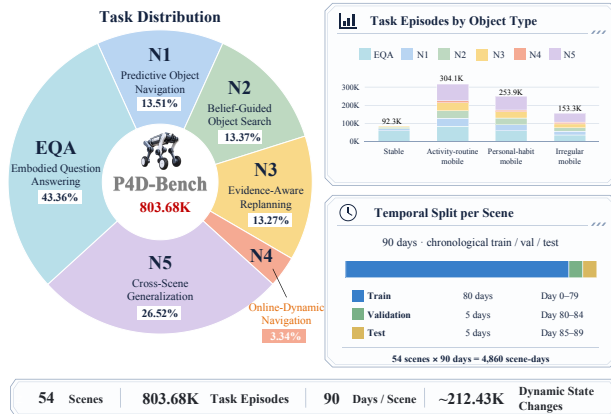


Figure 4: Overview of P4D-BENCH evaluation protocols.

Table 1: Mobility regimes in P4D-BENCH. Labels are used only for stratified evaluation and are not provided to the agent.

Regime	Mechanism	Frozen signal	Role
Stable/rare	Infrequent events	Home/return tendency	Persistence control
Routine	Activity state machine	Chain-specific locations	Routine prediction
Personal	Placed/carried lifecycle	Owner placement habit	Long-term memory
Irregular	Affordance sampling	Affordance only	Uncertainty control

and recovery. VLN supplies language-conditioned goals over the same evolving histories, while EQA probes whether memory preserves the evidence needed to reason about current and historical states. The controlled experiments in this paper emphasize state prediction and navigation, with EQA as a memory diagnostic; the remaining task annotations are included to support broader study of evolving-world agents.

4.2 HUMAN-TRACE-GROUNDED WORLD EVOLUTION

Source alignment. We derive complementary statistics from longitudinal smart-home records such as CASAS (Cook et al., 2013) and ARAS, and action-object evidence from EPIC-KITCHENS-100 and OPPORTUNITY. These sources are normalized into a common event schema linking time, activity, actor context, object category, source state, destination state, and evidence provenance. They constrain schedules, activity-object associations, transition timing, and placement tendencies; no source trajectory is copied verbatim into simulation. Unsupported quantities are explicitly marked as benchmark-authored rather than presented as measured human behavior.

Persistent household generation. Across 54 HSSD scenes (Khanna et al., 2024a) and multi-day household histories, we freeze household-level factors including resident schedules, room preferences, orderliness, object ownership, and placement habits. We then jointly generate activities, resident region trajectories, and object transitions in chronological order. Frozen factors make earlier observations informative about later states, whereas stochastic events, delayed returns, and exceptions prevent the histories from becoming deterministic scripts. The audited configuration yields 42.6 region transitions per resident-day, 6.56 relocations per personal object-day, 52.6% within-region relocations, and zero terminal collisions.

4.3 CONTROLLED DYNAMICS AND MOBILITY REGIMES

Every relocation is attributable to an activity, resident transition, object lifecycle event, or tidying event. Targets follow one of four complementary mechanisms shown in Table 1. Stable or rarely moved objects measure whether a predictive method preserves valid history. Routine objects follow repeated activity chains. Personal objects alternate between placed and carried states and exhibit owner-specific return habits. Irregular objects use affordance-valid destinations without temporal or actor-specific predictability. Mobility labels support stratified analysis but are hidden from the agent.

For each compatible history and query, we construct paired *static*, *routine*, and *random* worlds. Static preserves the last supported state; routine samples from the household’s frozen transition process; and random matches legal candidates and marginal movement statistics while breaking the learnable temporal dependency. Pair members share all public setup variables. Consequently, improvement specific to routine worlds is evidence that a model uses historical regularity rather than category frequency or spatial convenience. Audits additionally verify that personal objects do not simply track their owners and that actor or activity identity does not predict irregular destinations beyond affordance.

4.4 PHYSICAL REALIZATION AND CAUSAL OBSERVATION

Generated transitions are replayed chronologically in Habitat. Only semantically valid, collision-free placements with at least one reachable inspection viewpoint enter the benchmark. For every candidate, we record navigability, geodesic cost, camera frustum coverage, unoccluded target fraction, and

Table 2: Prediction and navigation protocols in P4D-BENCH.

Task	Decision signal	Primary measure
P Current-state prediction	Region/receptacle belief	$R@k$ / NLL
N1 Predictive navigation	Top-1 belief	First-Go SR
N2 Belief-guided search	Full belief + cost	Search SR / cost
N3 Evidence-aware replanning	Updated posterior	Recovery SR
N4 Online-dynamic navigation	Arrival-time belief	Online Recovery SR
N5 Cross-scene generalization	Any protocol above	Held-out SR

success viewpoints. These checks prevent a method from being penalized for an impossible placement or an unreachable observation.

A robot patrols using the map, time, travel cost, and its own accumulated history; it never observes current object states, resident locations, hidden activities, or generator latents. Region visits produce local RGB-D snapshots that are converted into positive and negative observations with visibility metadata. For a target queried at t_q , the public input terminates at or before t_q and contains the observed history, last supported state, elapsed time, calendar context, legal candidates, and spatial context. The evaluator retains the current state, hidden transitions, query-time activity, mobility regime, and oracle viewpoints. Query times cover both the natural temporal distribution and mechanism-relevant windows, such as stages of a routine or the interval after a personal object is dropped.

4.5 TASKS AND EVALUATION PROTOCOLS

P4D-BENCH derives all prediction and navigation episodes from the same target-time queries (Figure 4 and Table 2). The prediction track asks for a calibrated belief over the current region or receptacle without navigation confounds. A navigation episode augments that query with a start pose, candidate inspection viewpoints, a path budget, and frozen success conditions. N1 permits only the Top-1 predicted destination and measures whether prediction becomes a useful physical decision. N2 exposes the full belief and measures cost-aware search. N3 updates the belief after visibility-qualified observations and tests recovery from an incorrect first destination. N4 advances the world clock during execution and can introduce another hidden transition. N5 is not a separate task; it applies the preceding protocols to held-out scenes and households.

The EQA suite queries existence, location, temporal order, activity association, count, comparison, and compositional relations over the same causal memories. Answers are evaluated against a normalized ontology, and questions requiring future evidence are excluded. VLN episodes replace an object label with a language-conditioned goal whose referents and constraints are grounded in the same versioned scene history. Across tracks, methods receive identical public observations, candidates, geometry, starts, controllers, and budgets; only their memory representation, current-state belief, and resulting decisions differ.

4.6 SPLITS, LEAKAGE PREVENTION, AND QUALITY CONTROL

We split complete temporal sequences rather than individual queries so that no future portion of a household day can leak into an earlier decision. Held-out homes are reserved for cross-scene evaluation, and paired static/routine/random members remain in the same split. Near-duplicate queries and transition groups are assigned atomically. The serialized release is audited for future timestamps, private event fields, answer-correlated filenames or indices, repeated event groups across splits, inconsistent paired variants, invalid physics, unreachable goals, and accidental dependence of the patrol policy on hidden state.

Results are reported by mobility regime, transition class, staleness, spatial difficulty, and generalization split. We also expose dataset cards and audit summaries for source mappings, authored assumptions, physical validity, candidate coverage, and leakage tests. Full schema definitions, episode validity rules, and metric implementations are provided in the supplement.

5 EXPERIMENTS

We evaluate three linked questions: whether P4D-NAV predicts the latent current state more accurately and with better calibration; whether its belief improves navigation destinations and search efficiency; and whether the closed-loop agent recovers when its initial prediction conflicts with physical evidence. Benchmark experiments follow the native protocols described below; within each comparison, methods share perception, candidate states, inspection viewpoints, low-level control, and action budgets.

5.1 EVALUATION MATRIX

Cross-benchmark evaluation. We evaluate the complete agent on P4D-BENCH, FindingDory (Yadav et al., 2025), and GOAT-Bench (Khanna et al., 2024b). The three benchmarks probe complementary capabilities rather than sharing a single pooled score. P4D-BENCH isolates belief over hidden current state and its effect on evidence-aware navigation. FindingDory evaluates long-horizon memory through high-level goal selection and conditional low-level execution; the compact comparison reports HL-SR, LL-SR, and HL-SPL, while the appendix retains the full high- and low-level breakdown. GOAT-Bench evaluates lifelong multi-goal navigation; in addition to its official SR and SPL, we report Repeat-SR on subtasks whose target identity or containing region has appeared in the agent’s prior stream. The latter is an explicitly marked diagnostic of memory reuse, not an official leaderboard metric. Category, language, and image goals are disaggregated in the appendix. On P4D-BENCH, First-Go SR, Search SR, and SPL jointly expose prediction quality, eventual task success, and path efficiency. Every comparison uses the benchmark’s native protocol, and reported external numbers are separated from reproduced results.

Cross-domain simulation. To distinguish temporal reasoning from renderer- or dataset-specific fitting, we evaluate the same system in conventional mesh-based environments (MP3D and HM3D) and high-fidelity Gaussian-splatting environments (Habitat-GS and Interior-GS). Scene-specific candidate sets and navigable viewpoints are constructed with the same rules, while perception and control configurations are held fixed wherever simulator interfaces permit. We report both in-domain performance and transfer from the P4D-HSSD training distribution.

Real-robot evaluation. We further evaluate DR-AGENTOS in a real indoor office–laboratory environment using three commercial legged robotic platforms: DEEP Robotics LYNX M20, X30, and Lite3. M20 serves as the primary platform for quantitative evaluation, while X30 and Lite3 are used to examine whether the same predictive-memory and belief-guided search framework transfers across different robot platforms. The memory representation, prediction module, belief update, and high-level search policy are shared across platforms; only the robot-specific perception and low-level navigation interfaces differ. Detailed manufacturer-reported platform specifications are provided in the appendix.

The real-world evaluation focuses on budgeted object search under persistent indoor changes. Target objects are observed over multiple sessions and may be relocated between observations according to either recurring routines or routine-breaking changes. At query time, the agent must infer the target’s current location from stale history, select candidate locations to inspect, navigate to them, and update its belief when physical observations contradict the initial prediction. Targets remain stationary during each search episode, so the evaluation isolates hidden state changes between observations rather than target motion during execution.

Compared methods receive the same pre-query history, candidate locations, starting condition, perception interface, and platform-specific low-level navigation stack. Success requires autonomous online confirmation of the target within the search budget; missed detections, navigation failures, timeouts, and human interventions count as failures. We report Search SR, First-Go SR, travel distance, elapsed time, number of inspected locations, and recovery after an incorrect first prediction. Corresponding outcomes and failure analysis are reported in Section 6.6.

5.2 BASELINES AND ABLATIONS

We select baselines from the implementations that have been validated in our evaluation stack and organize them by the capability they contribute. The *open-vocabulary and lifelong navigation* group contains VLFM (Yokoyama et al., 2024), ZSON (Majumdar et al., 2022), the FindingDory agent (Yadav et al., 2025), and LagMemo GLUE (Zhou et al., 2025). These methods test frontier-based semantic search, learned zero-shot navigation, long-horizon episodic recall, and persistent 3DGS language memory, respectively. The *structured spatio-temporal memory* group contains HOV-SG (Werby et al., 2024) and DynaMem (Liu et al., 2024), representing hierarchical open-vocabulary graphs and online dynamic metric-semantic memory. The *predictive world-model* group contains SLaTe-PRO (Patel et al., 2023), SGM+NEP (Kurenkov et al., 2023), FlowMaps (Argenziano et al., 2026), and the Perpetua predictor from PredictiveGraphs (Saavedra-Ruiz et al., 2026).

Navigation systems are executed through their released policies or benchmark adapters. Memory-only methods replace DR-AGENTOS’ memory backend while using the same detector, goal interface, and low-level controller. Prediction-only methods receive the same causal history and candidate set; their state scores are passed to the same inspection planner. This component-aligned adaptation is reported separately from native published results and prevents a stronger detector or controller from being mistaken for better temporal reasoning. Last Seen, Frequency, Markov, and Random Forest remain lightweight diagnostic baselines in the dedicated P4D-BENCH and predictor tables rather than the headline cross-benchmark comparison.

N4 additionally compares against Transit-Aware Planning under its required online target-path input (Dorbala et al., 2026). Oracle Current State provides the navigation upper bound. Oracle Activity supplies the ground-truth current activity and is reported only as a diagnostic rather than a fair baseline.

For EQA, all methods use the same question templates, answer ontology, and normalization. Queried information must be recovered from the memory available to each method. When a learned language readout is required, it is held fixed across memory methods so that the comparison primarily measures memory quality rather than language-model capacity.

The principal ablations remove continuous-time features; the persistence–relocation factorization; instance-specific history; calendar context; negative evidence; visibility calibration; Bayesian belief updating; or the full distribution in favor of Top-1 prediction. *P4D-Belief Prior* predicts once and searches without posterior updates, while full P4D-NAV closes the observe–update–replan loop.

We additionally replace the predictive component with Frequency, Markov, Random Forest, GRU, direct Transformer, and the structured continuous-time Transformer while keeping memory, candidates, and the agent policy fixed. To test whether the system depends on one proprietary controller, we run the same typed memory and navigation tools with GPT-family and Qwen-VL-family foundation models, plus a third VLM where available. Comparisons require fixed model versions, prompts, temperature, tool schemas, and inference budgets; these configuration fields belong to the reproducibility record in Appendix A.7.

5.3 METRICS AND REPORTING

We first assess predictive belief independently of navigation using state Top-1/Top- k accuracy, negative log-likelihood (NLL), Brier score, expected calibration error (ECE), true-state rank, mean reciprocal rank (MRR), and persistence AUROC.

For navigation, we report Success Rate (SR), SPL, path length, completion time, First-Go SR, number of inspected candidates, and search cost. N3 additionally reports Recovery SR after an incorrect first prediction and posterior true-state rank improvement after negative evidence. N4 reports Online Recovery SR, replanning count, replanning delay, and excess path length relative to a trajectory-aware oracle. This oracle is required because motion during execution makes the conventional shortest path to a fixed terminal goal ill-defined.

For EQA, we report answer accuracy by existence, location, temporal, activity, count, comparison, and reasoning question, together with category-macro accuracy. Structured outputs are normalized to the benchmark ontology before scoring.

Table 3: Cross-benchmark comparison grouped by method capability. FindingDory reports high-level success (HL-SR), low-level success (LL-SR), and high-level SPL (HL-SPL); GOAT-Bench reports SR, SPL, and repeated-goal success (Repeat-SR); P4D-BENCH reports first-destination success (First-Go SR), eventual search success (Search SR), and SPL. Repeat-SR is our diagnostic; all other columns use the corresponding benchmark protocol. Dashes denote unavailable measurements.

Method	FindingDory			GOAT-Bench			P4D-BENCH		
	HL-SR ↑	LL-SR ↑	HL-SPL ↑	SR ↑	SPL ↑	Repeat-SR ↑	First-Go SR ↑	Search SR ↑	SPL ↑
<i>Open-vocabulary and lifelong navigation</i>									
VLFM (Yokoyama et al., 2024)	–	–	–	–	–	–	–	–	–
ZSON (Majumdar et al., 2022)	–	–	–	–	–	–	–	–	–
FindingDory Agent (Yadav et al., 2025)	–	–	–	–	–	–	–	–	–
LagMemo GLUE (Zhou et al., 2025)	–	–	–	–	–	–	–	–	–
<i>Structured spatio-temporal memory</i>									
HOV-SG (Werby et al., 2024)	–	–	–	–	–	–	–	–	–
DynaMem (Liu et al., 2024)	–	–	–	–	–	–	–	–	–
<i>Predictive world and state modeling</i>									
SLaTe-PRO (Patel et al., 2023)	–	–	–	–	–	–	–	–	–
SGM+NEP (Kurenkov et al., 2023)	–	–	–	–	–	–	–	–	–
FlowMaps (Argenziano et al., 2026)	–	–	–	–	–	–	–	–	–
Perpetua (Saavedra-Ruiz et al., 2026)	–	–	–	–	–	–	–	–	–
DR-AGENTOS (ours)	–	–	–	–	–	–	–	–	–

Benchmark aggregate results include bootstrap confidence intervals over episodes and both micro and mobility-regime macro averages. We stratify results by paired world variant, mobility regime, transition class (*stay*, *same-region move*, *cross-region move*, *multi-move*, and *return*), staleness, spatial difficulty, grounding quality, and generalization split. The key causal contrast is the paired routine gain: improvement in the routine world relative to performance on its paired static and random variants. This contrast tests whether gains arise from persistent temporal regularities rather than scene geometry, identity, or generic search behavior. Implementation and reporting details appear in Appendices A.7 and A.8.

6 RESULTS AND ANALYSIS

The evaluation separates six questions: performance across embodied benchmarks, predictive navigation on P4D-BENCH, visual-domain generalization, the role of individual components, controller dependence, and indoor/outdoor object search. *Draft data status*: – denotes an unavailable measurement. The real-world table contains explicitly marked numerical placeholders; its associated analysis is provisional and must be recomputed from robot outcomes before submission.

6.1 COMPARISON ACROSS EMBODIED BENCHMARKS

The groups mirror the source of each method’s advantage rather than suggesting that benchmark scores are interchangeable. Released navigation systems are run through their native policies when possible; memory and prediction methods are inserted into a shared controller as specified in Section 5.2. Appendix tables further separate FindingDory high-/low-level execution and GOAT category/language/image goals so that an aggregate score cannot hide a modality-specific failure.

6.2 PREDICTIVE NAVIGATION ON P4D-BENCH

First-Go SR isolates whether a better current-state belief becomes a better physical destination. Search SR and cost then test whether probability mass beyond Top-1 is useful. Recovery SR is conditioned on an incorrect first prediction so that easy first-attempt successes do not mask poor evidence use. The relevant estimands are paired episode-level differences with confidence intervals that retain the pairing.

Qualitative case study. Figure 5 traces one routine-world episode from stale memory to physical verification. The target bowl was last observed in a cabinet 14.77 hours before the query, but a dinner-cleanup transition relocates it to the kitchen sink. The predictive belief ranks the sink first, allowing P4D-NAV to verify the target in one inspection; the comparison routes expose how an obsolete last-seen state or a geometry-only ordering increases search cost.

Table 4: Main results on routine-world N1–N3 episodes. All methods share perception, candidates, inspection viewpoints, and the low-level navigator. Lower is better for inspections and distance.

Method	First-Go SR ↑	Search SR ↑	SPL ↑	Inspect. ↓	Distance ↓	Recovery SR ↑
<i>Navigation and persistent memory</i>						
VLFM (Yokoyama et al., 2024)	–	–	–	–	–	–
FindingDory Agent (Yadav et al., 2025)	–	–	–	–	–	–
LagMemo GLUE (Zhou et al., 2025)	–	–	–	–	–	–
HOV-SG (Werby et al., 2024)	–	–	–	–	–	–
DynaMem (Liu et al., 2024)	–	–	–	–	–	–
<i>Predictive world and state modeling</i>						
SLaTe-PRO (Patel et al., 2023)	–	–	–	–	–	–
SGM+NEP (Kurenkov et al., 2023)	–	–	–	–	–	–
FlowMaps (Argenziano et al., 2026)	–	–	–	–	–	–
Perpetua (Saavedra-Ruiz et al., 2026)	–	–	–	–	–	–
P4D-Nav (ours)	–	–	–	–	–	–
Oracle Current State	–	–	–	–	–	–

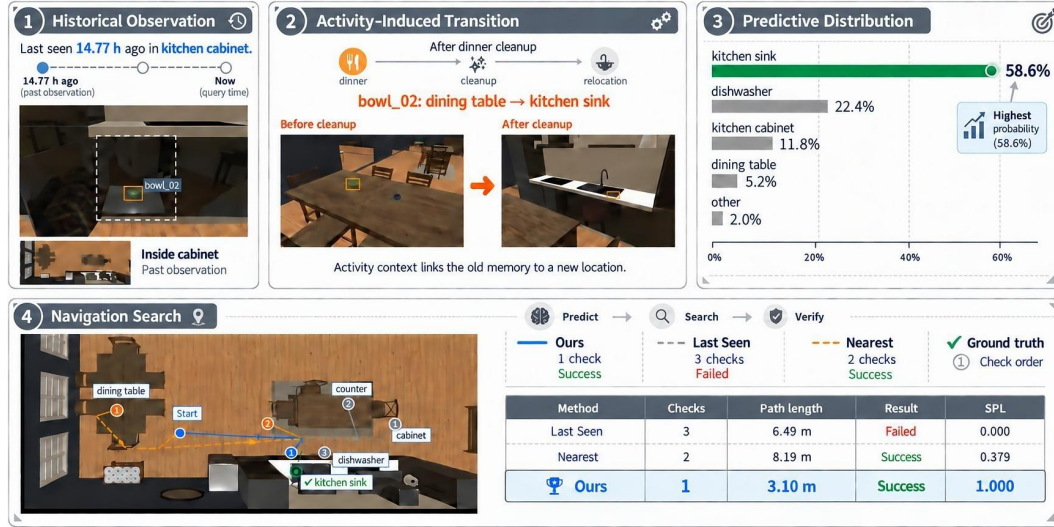


Figure 5: P4D-Bench case study from historical observation to belief-guided navigation. Predictive search reaches the activity-induced current state in one inspection, while last-seen and nearest-first search follow less efficient candidate orders.

Temporal and budget dependence. Memory staleness, inspection budget, and mobility regime define complementary axes of analysis. SR as a function of staleness tests robustness to observation gaps; SR as a function of inspection budget tests the value of alternatives beyond Top-1; and paired routine gains separate learnable temporal structure from matched random transitions. These diagnostics require episode-level measurements and are not inferred from aggregate SR.

6.3 GENERALIZATION ACROSS SIMULATION FIDELITY

Table 5: Generalization across conventional mesh-based and high-fidelity Gaussian-splatting environments. Methods use released weights or the fixed source-domain training split and are evaluated without target-domain fine-tuning; SR and SPL use matched task and budget definitions.

Method	MP3D		HM3D		Habitat-GS		Interior-GS	
	SR	SPL	SR	SPL	SR	SPL	SR	SPL
<i>Open-vocabulary and lifelong navigation</i>								
VLFM	-	-	-	-	-	-	-	-
ZSON	-	-	-	-	-	-	-	-
LagMemo GLUE	-	-	-	-	-	-	-	-
<i>Structured spatio-temporal memory</i>								
HOV-SG	-	-	-	-	-	-	-	-
DynaMem	-	-	-	-	-	-	-	-
<i>Predictive world and state modeling</i>								
FlowMaps	-	-	-	-	-	-	-	-
Perpetua	-	-	-	-	-	-	-	-
DR-AGENTOS (ours)	-	-	-	-	-	-	-	-

The principal comparison is not only average SR but the degradation from mesh-based to high-fidelity rendering. A breakdown into belief, grounding, detection, and navigation errors distinguishes shifts in temporal prediction from shifts in visual evidence.

6.4 WHAT COMPONENTS PRODUCE THE GAIN?

Table 6: Module-level ablation. Prediction metrics test belief quality; agent metrics test whether each component changes useful behavior.

Variant	Top-1	NLL	SR	SPL	Inspect.
w/o predictive belief	-	-	-	-	-
w/o 4D memory (latest observation)	-	-	-	-	-
w/o continuous-time encoding	-	-	-	-	-
w/o persistence-relocation head	-	-	-	-	-
w/o visibility calibration	-	-	-	-	-
w/o Bayesian evidence update	-	-	-	-	-
Full DR-AGENTOS	-	-	-	-	-

The two top-level removals test the paper’s central system claims: whether prediction adds value beyond retrospective memory, and whether temporally versioned evidence adds value beyond the latest observation. The remaining rows localize gains within the predictor and agent loop. Joint interpretation of NLL and navigation cost distinguishes improved classification from improved decisions.

Table 7: Predictor comparison with all non-predictive modules fixed. Runtime is measured per target query on the same hardware.

Predictor	Top-1	NLL	ECE	First-Go SR	Latency
Frequency	-	-	-	-	-
Markov	-	-	-	-	-
Random Forest	-	-	-	-	-
GRU	-	-	-	-	-
Transformer Direct	-	-	-	-	-
Structured CT-Transformer	-	-	-	-	-

6.5 FOUNDATION-MODEL ADAPTATION

Table 8: Compatibility with different frozen foundation-model controllers. All models receive identical typed tools, retrieved evidence, prompts, and budgets.

Controller	Tool valid	Task SR	SPL	EQA Acc.	Latency
GPT-family VLM	–	–	–	–	–
Qwen-VL-family	–	–	–	–	–
Third frozen VLM	–	–	–	–	–

This experiment tests interface portability, not which foundation model is universally strongest. Tool-valid rate measures whether a controller follows the same structured action contract; the remaining columns measure whether the core benefit of predictive memory persists across model families.

6.6 REAL-WORLD EXPERIMENTS

We organize the indoor and outdoor comparison around two questions: whether history-informed search increases budgeted success, and whether its decisions reduce the motion and time needed to inspect candidate locations. Table 9 presents the complete comparison layout. The provisional values come exclusively from the first dataset in the supplied report; multi-replicate averages and sensitivity scenarios are excluded from this table.

Table 9: Indoor/outdoor search comparison—**draft numerical placeholders, not physical measurements**. Values reproduce the first synthetic dataset in `RESULTS.md`, with 32 samples per method and environment. The four method slots use Last-seen rule, Time-frequency rule, Retrieval proxy, and Predictive proxy values, respectively; they do not establish performance of the named robot implementations. Replace all entries with measured robot outcomes. Distance and time are means over all attempts, including failures.

Environment	Method slot	n	Successes	SR (%)	Distance (m)	Time (s)
Indoor	Last Seen + Search	32	13	40.62	17.15	65.73
	Time Frequency	32	14	43.75	20.14	75.26
	Retrieval + Reasoning	32	17	53.12	18.29	96.53
	DR-AGENTOS	32	16	50.00	20.23	74.56
Outdoor	Last Seen + Search	32	13	40.62	45.19	111.67
	Time Frequency	32	13	40.62	43.87	109.13
	Retrieval + Reasoning	32	13	40.62	38.92	128.51
	DR-AGENTOS	32	12	37.50	42.22	106.37

Indoor search. In the provisional comparison, the DR-AGENTOS slot reaches 16/32 successes (50.00%), compared with 14/32 for Time Frequency and 13/32 for Last Seen + Search. The corresponding differences are +6.25 and +9.38 percentage points, computed from counts before rounding. Retrieval + Reasoning occupies the highest-success slot at 17/32 (53.12%). The DR-AGENTOS slot has a mean elapsed time of 74.56 seconds, compared with 96.53 seconds for Retrieval + Reasoning, but travels farther (20.23 versus 18.29 meters). Thus, the numerical example illustrates a success–time tradeoff, rather than uniform dominance in success and travel cost.

Outdoor search. The outdoor placeholders give 12/32 successes (37.50%) for the DR-AGENTOS slot and 13/32 (40.62%) for each comparison method. Its mean elapsed time is 106.37 seconds, compared with 109.13 seconds for Time Frequency and 128.51 seconds for Retrieval + Reasoning. Its travel distance is lower than Time Frequency’s (42.22 versus 43.87 meters) but higher than Retrieval + Reasoning’s (38.92 meters). A shorter attempt at lower SR does not establish more efficient successful search: early failures can reduce both elapsed time and distance.

Aggregate contrast and uncertainty. The primary contrast is the equally weighted mean of the two within-environment SR differences between DR-AGENTOS and Time Frequency. In the placeholder counts, this is $(6.25 - 3.125)/2 = 1.5625$ percentage points, corresponding to 28/64 versus 27/64 successes. Pooling the equal-sized environments gives the same result but obscures the reversal in the within-environment comparison. These descriptive differences do not establish a statistically

reliable gain. Physical paired uncertainty requires the actual block outcomes and object/day identities; quantiles across synthetic studies cannot supply that uncertainty.

Evidence use and failure analysis. Aggregate SR and attempt costs do not determine whether a failure arose from an incorrect belief, poor visibility, missed detection, or navigation error. Likewise, they do not establish the contribution of posterior updating or continuous-time prediction. First-inspection confirmation, recovery conditional on a wrong initial choice, and the history/action ablations therefore require separate run-level evidence. A recovery case links the causal history and initial belief to an executed inspection, its sensor evidence, the posterior, and the next action. No physical case trace or recovery rate is inferred from the placeholder table. Section 5.1 defines the corresponding measurement protocol.

7 CONCLUSION

We introduced persistent navigation in evolving worlds, a setting in which an embodied agent must act on a long-term but potentially stale account of its environment. DR-AGENTOS provides a memory-belief-action loop for inferring task-relevant current state, while its P4D-NAV core operationalizes this problem through dynamic object goals: it separates persistence from relocation, represents variable scene-specific candidates, converts visibility-qualified observations into posterior evidence, and uses the evolving distribution to navigate, explore, and replan. P4D-BENCH grounds this question in human-trace-driven HSSD histories and paired static, routine, and random worlds spanning prediction, navigation, and embodied reasoning. The resulting formulation reframes navigation from executing against a remembered snapshot to acting on the world state best supported by accumulated and newly collected evidence. Quantitative conclusions will be added only after the multi-scene navigation evaluation is complete.

8 LIMITATIONS

Although DR-AGENTOS is motivated by general task-relevant world evolution, the current predictive model is operationalized primarily through discrete portable-object states; structural reconfiguration and changing traversability require richer state and action models. P4D-NAV also assumes reliable entity association and an available set of candidate receptacles and inspection viewpoints; pose drift, missed detections, identity switches, or missing candidates can corrupt both prediction and navigation. Its discrete belief represents arbitrary continuous placements less directly than continuous generative models. The source datasets are complementary rather than jointly observed, so P4D-BENCH episodes are human-trace-grounded simulations, not reconstructions of one household. The main protocol freezes the target after the query; N4 is required to test simultaneous target and agent motion. Finally, visibility-aware updates reduce weak negative evidence but cannot guarantee recovery when beliefs and perception fail together or routines shift.

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A SUPPLEMENTARY MATERIAL

A.1 FORMAL EPISODE INTERFACE

Each navigation episode references one causal prediction query. Public fields include schema version, episode and query identifiers, split, task type, scene and household identifiers, world variant, target description, query time, agent start pose, candidate-state identifiers, budgets, and success criteria. Evaluation-private fields include the current state, target pose, valid goal viewpoints, hidden transition class, oracle path, mobility label, and query-time activity. Train and validation releases may expose analysis labels in metadata; the formal test release replaces them with null values and computes grouped metrics in a private evaluator.

The high-level track exposes

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NAVIGATE_TO(state), INSPECT(state), STOP, NOT_FOUND,
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with a fixed navigation backend. The low-level track exposes Habitat-style forward, turn, look, and stop actions. The main paper uses the high-level track to isolate predictive state selection; low-level results measure sensitivity to control and perception.

A.2 EPISODE VALIDITY AND LEAKAGE AUDIT

An episode is retained only if: (i) the target has at least one positive pre-query observation; (ii) the target state is known to the private evaluator and physically instantiated without collision; (iii) at least one valid viewpoint lies on the NavMesh; (iv) the start is on the NavMesh, at least 3 meters from the target, and does not reveal it; and (v) all public histories terminate at or before the query. A successful stop must be within 1 meter geodesic distance of a valid viewpoint, observe the target above a frozen visible-fraction threshold, and identify the correct instance or category.

We audit serialized examples for future timestamps, hidden event fields, filenames or indices correlated with answers, duplicate event groups across splits, and inconsistent paired variants. Static, routine, and random members of a pair must have identical public query fields. Invalid simulator episodes are reported separately and are not counted as ordinary failures.

A.3 MOBILITY-TYPE ASSIGNMENT

The frozen rule is

$$\text{type}(o) = \begin{cases} \text{stable}, & r_{\text{move}}(o) < \tau_{\text{move}}, \\ \text{activity}, & G_{\text{act}}(o) \geq \tau_{\text{act}}, \\ \text{personal}, & C_{\text{habit}}(o) \geq \tau_{\text{habit}}, \\ \text{irregular}, & \text{otherwise.} \end{cases} \quad (16)$$

All thresholds are selected on training data before navigation evaluation. When the generator specifies a mobility profile, the same statistics are used to verify that realized trajectories exhibit the intended behavior. Every group must cover several categories, instances, and scenes, and no category may uniquely reveal one mobility type.

A.4 METRIC DEFINITIONS

Let S_i be episode success, L_i the executed path, and L_i^* the oracle shortest path to the query-time goal under the same start. Standard SPL is

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{L_i^*}{\max(L_i, L_i^*)}. \quad (17)$$

First-Go SR counts success when only the initial predicted candidate may be inspected. Recovery SR conditions on an incorrect first candidate and measures whether the target is subsequently found. Excess path is $L_i - L_i^*$ for static-after-query episodes. For online-dynamic episodes, the oracle is trajectory aware and may intercept the target; we therefore report both success within the common budget and $L_i - L_{i,\text{dyn}}^*$.

Calibration metrics operate on the full candidate belief before navigation and after each update. ECE bins maximum confidence, Brier score evaluates all candidates, and true-state rank avoids hiding useful changes below Top-1. Search-cost curves plot success against allowed inspections and travelled distance rather than reporting one arbitrary budget.

A.5 FULL BASELINE SPECIFICATION

Last Seen assigns all mass to the latest positive state. Frequency Prior estimates $P(s \mid o)$ from training data. Markov Transition estimates $P(s_{k+1} \mid s_k, o)$; Time-Conditioned Prior additionally conditions on public hour and weekday bins. Instance Hotspot uses only the target instance’s training trajectory. GRU Direct and Transformer Direct receive the same event tokens and candidate encoder as P4D-NAV but directly normalize candidate logits without the persistence factorization. P4D-Belief Prior uses Equation (11) once at episode start. Full P4D-NAV applies Equation (14) and Equation (12) after each inspection.

The Oracle Activity diagnostic may use the hidden activity and source location but is excluded from fair rankings. Oracle Current State receives the private target state and measures remaining perception and navigation error. External predictive methods are adapted only through their public inputs; any use of privileged simulator state must be identified and excluded from the main comparison.

A.6 DISAGGREGATED RESULT TEMPLATES

Table 10: FindingDory results using its native high- and low-level metrics. The main paper reports HL-SR, LL-SR, and HL-SPL for compactness.

Method	HL-SR	HL-SPL	LL-SR	LL-SPL
FindingDory Agent	–	–	–	–
LagMemo GLUE	–	–	–	–
HOV-SG	–	–	–	–
DynaMem	–	–	–	–
DR-AGENTOS	–	–	–	–

Table 11: GOAT-Bench modality-wise results on the official evaluation split. The main paper reports aggregate SR, SPL, and the Repeat-SR memory diagnostic.

Method	Category goal		Language goal		Image goal	
	SR	SPL	SR	SPL	SR	SPL
VLFM	–	–	–	–	–	–
ZSON	–	–	–	–	–	–
FindingDory Agent	–	–	–	–	–	–
LagMemo GLUE	–	–	–	–	–	–
DR-AGENTOS	–	–	–	–	–	–

Table 12: Search SR by hidden mobility type. Types are determined from training history and withheld from the agent. Macro average weights the four groups equally.

Method	Stable	Activity routine	Personal habit	Irregular	Macro
Last Seen	–	–	–	–	–
Time-Conditioned	–	–	–	–	–
Instance Hotspot	–	–	–	–	–
Transformer Direct	–	–	–	–	–
P4D-NAV	–	–	–	–	–

A.7 IMPLEMENTATION DETAILS

Unless otherwise noted, the history encoder has three Transformer layers, hidden width 128, four attention heads, and at most 64 events. The predictive model contains approximately –million trainable parameters and is optimized with AdamW using validation NLL for checkpoint selection. Candidate encoders are shared across scenes. The main high-level track fixes candidate viewpoints and low-level planning; a Habitat low-level action track evaluates the complete embodied stack. Final learning rate, batch size, epoch count, hardware, simulator version, detector, visibility thresholds, and repeated-run protocol will be frozen before test evaluation.

A.8 REPRODUCIBILITY CHECKLIST

The completed release will report:

- source dataset versions, licenses, mappings, filters, and canonical-record statistics;
- HSSD and Habitat versions, scene identifiers, object assets, placement checks, and NavMesh settings;
- split manifests, paired-example identifiers, checksums, and leakage-audit output;
- model width, layers, attention heads, time features, history length, optimizer, learning rate, epochs, seeds, and checkpoint rule;
- detector and depth inputs, visibility computation, candidate and action budgets, evidence likelihoods, and planner hyperparameters;
- hardware, wall-clock latency, memory growth, repeated-run aggregation, and confidence intervals; and

Table 13: Paired-world navigation results. Variants share all observed setup variables and differ only in hidden transitions.

Method	Static SR	Routine SR	Random SR	Routine gain over Last Seen
Last Seen	—	—	—	—
Markov Transition	—	—	—	—
Transformer Direct	—	—	—	—
P4D-NAV	—	—	—	—

- reproduced versus quoted baseline values, grouped results, failure labels, and all component-aligned ablations.

A.9 SUBMISSION COMPLIANCE NOTES

This source is configured for anonymous ICLR 2027 submission: \iclrfinalcopy remains disabled, author identities and affiliations are absent, and references precede the appendix. Before submission, authors should remove all result placeholders, keep the main text within the conference page limit, and scan figure metadata, linked repositories, PDF properties, acknowledgements, and supplementary files for identifying information.

A.10 ETHICS, REPRODUCIBILITY, AND AI USE

Ethics. Persistent visual memory may capture people or private spaces, while an incorrect belief may induce unnecessary motion or unsafe behavior. Deployments should obtain consent, enforce retention and access controls, preserve provenance, and retain platform-specific collision avoidance and emergency stops outside the learned model. Our benchmark compiles existing records into simulated HSSD episodes and introduces no new human-subject study.

Reproducibility. Section 3 specifies the task, memory, predictor, evidence update, and agent policy. Section 4 defines episode construction and leakage controls, while Section 5 specifies baselines and metrics. An anonymized code and data snapshot will accompany the review submission.

AI use. Generative AI tools assisted manuscript editing, LaTeX checks, and local engineering. They did not generate benchmark observations, ground-truth labels, or reported scores. The authors reviewed all assisted material and remain responsible for the content and artifacts.